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Nomenclature

Physical constants

ε_0	Permittivity of free space	
μ_0	Permeability of free space	
a_0	Bohr radius	
c	Speed of light in vaccuum	
e	Elementary charge	
h	Planck constant	
k_B	Boltzmann constant	
m_e	Electron mass	
N_A	Avogadro's number	
$\mathcal{R}$	Rydberg energy	

Scalars

α	Attenuation coefficient	m^{-1}
$\dot{\alpha}$	Volume fraction rate of change	s^{-1}
β	Thermoelectric coefficient	1
Γ	Reaction rate density	$\mathrm{m}^{-3}\,\mathrm{s}^{-1}$
γ	Adiabatic index	1
ε_r	Relative permittivity	1
η	Refractive index	1
κ	Thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
μ	Dynamic viscosity	$\mathrm{Pa\,s}$
μ_r	Relative permeability	1

ν	Frequency	Hz
ρ	Charge density	$\mathrm{C\,m^{-3}}$
ρ	Mass density	$\mathrm{kg\,m^{-3}}$
σ	Reaction cross section	$\mathrm{m^2}$
Φ	Quantum yield	1
ϕ	Scalar flux	$\mathrm{m^{-2}\,s^{-1}}$
χ	Magnetization	1
ω	Angular frequency	$\mathrm{rad\,s^{-1}}$
A	Area	$\mathrm{m^2}$
$\mathcal{E}$	Particle internal energy	J
ρE	Internal energy density	$\mathrm{J\,m^{-3}}$
E	Specific internal energy	$\mathrm{J\,kg^{-1}}$
$\mathcal{H}$	Particle enthalpy	J
ρH	Enthalpy density	$\mathrm{J\,m^{-3}}$
H	Specific enthalpy	$\mathrm{J\,kg^{-1}}$
I	Intensity	$\mathrm{W\,m^{-2}}$
M	Molar mass	$\mathrm{kg\,mol^{-1}}$
m	Mass	kg
N	Number	1
n	Number density	$\mathrm{m^{-3}}$
p	Pressure	Pa
q'''	Volumetric heat generation rate	$\mathrm{W\,m^{-3}}$
S	Reaction rate coefficient	varies
s	Distance	m
T	Temperature	K
t	Time	s
$\mathcal{U}$	Particle potential energy	J
V	Volume	$\mathrm{m^3}$

v	Speed	$\mathrm{m\,s^{-1}}$
Vectors		
$\hat{\boldsymbol{\omega}}$	Direction	1
$\mathbf{a}$	Acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{B}$	Magnetic field	T
$\mathbf{E}$	Electric field	$\mathrm{V\,m^{-1}}$
$\mathbf{f}$	Force density	$\mathrm{N\,m^{-3}}$
$\mathbf{g}$	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{j}$	Current density	$\mathrm{A\,m^{-2}}$
$\hat{\mathbf{n}}$	Normal direction	1
$\mathbf{q}''$	Heat flux density	$\mathrm{W\,m^{-2}}$
$\mathbf{r}$	Position	m
$\mathbf{S}$	Poynting vector	$\mathrm{W\,m^{-2}}$
$\mathbf{u}$	Velocity	$\mathrm{m\,s^{-1}}$
$\rho\mathbf{u}$	Momentum density	$\mathrm{N\,s\,m^{-3}}$
Matrices		
$\mathbf{K}$	Conductivity tensor	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\mathbf{\Pi}$	Thermoelectric flux tensor	V
$\mathbf{T}$	Deviatoric stress tensor	Pa
Subscripts		
ci	Collisional ionization	
ib	Inverse bremsstrahlung	
pi	Photoionization	
rec	Recombination	
γ	Photon	
b	Bound electron	
e	Free electron	
i	Ion	
Superscripts		
K	Kinetic energy	
M	Momentum	

Chapter 1

Introduction

Chapter 2

Physics of laser-droplet interactions

Modeling the interaction between a laser beam and droplet combines the approaches of multiphase flow and magnetohydrodynamics (MHD). The former is simply due to the existence of the two immiscible liquid and gas phases, and the latter is especially necessary if the laser intensity is strong enough to ionize. In this paper, the term gas phase includes both neutral and charged particles, as both are governed by the ideal gas equation of state.

In order to accurately capture the behavior of the liquid-gas-plasma, a four-fluid model is used to describe the constituent fluids – namely, liquid, gas, electrons and ions – that interact with one another. The charged particles also exhibit electromagnetic effects. The governing equations are therefore derived from the three conservation laws and Maxwell's equations.

Chemically speaking, the laser can depart enough energies to not only remove bound electrons from their orbits but also break molecular bonds and alter the chemical composition of the fluid. In fact, free radicals and light molecular ions are frequently encountered in plasmas if the laser is not strong enough to completely disintegrate molecules into monoatomic ions. However, for simplicity, all ions are assumed to have resulted from neutral molecules each losing a single electron, and thus they have a charge of +1 and the same molecular weight as the parent neutral particle.

2.1 Fluid description of plasmas

2.1.1 Conservation equations

The multiphase flow model is based on the Baer-Nunziato model of a liquid-vapor system not in equilibrium. The model allows for distinct velocities, temperatures, and pressures between the phases and employs relaxation terms to drive those quantities into equilibrium. To extend this model to include the plasma state, it is first assumed that charged particles will only occupy the gas phase and interact solely with the neutral gas particles. The gas phase then acts the intermediate phase between liquid and plasma, and its flow equations have to include interactions (phase change and relaxations) with both. The relaxation terms between fluids j and k in the volume fraction, momentum, and energy equations are denoted $\dot{\alpha}_{jk}$, $\mathbf{f}_{jk}$, q'''_{jk}, respectively.

There are four fluids in this system, and the subscripts (and sometimes superscript) to identify them are l for liquid, g for neutral gas, e for electrons, and i for ions. One exception exists for particle mass m (a single particle does not exhibit state behavior), in which case m_e stands for electron mass and m stands for the neutral particle or ion mass. For laser interaction term, γ is also used to indicate photons (see Section 2.3). This convention is used for the rest of the document.

The conservation equations for the liquid phase includes the mass, momentum, and energy equations on top of an additional equation for the liquid volume fraction, α_l.

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot \alpha_l \mathbf{u}_l = \dot{\alpha}_{gl} \tag{2.1a}$$
$$\frac{\partial \alpha_l \rho_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l = 0 \tag{2.1b}$$
$$\frac{\partial \alpha_l \rho_l \mathbf{u}_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l \otimes \mathbf{u}_l = -\nabla p_l + \nabla \cdot \mathbf{T}_l + \mathbf{f}_{gl} \tag{2.1c}$$
$$\frac{\partial \alpha_l \rho_l E_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l H_l = -\nabla \cdot \mathbf{q}_l'' + \nabla \cdot [\mathbf{T}_l \cdot \mathbf{u}_l] + q_{gl}'''. \tag{2.1d}$$

The neutral gas equations are written similarly, but the gas volume fraction α_g can be computed algebraically as $\alpha_g = 1 - \alpha_l$:

$$\frac{\partial \alpha_g \rho_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g = 0 \tag{2.2a}$$
$$\frac{\partial \alpha_g \rho_g \mathbf{u}_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g \otimes \mathbf{u}_g = -\nabla p_g + \nabla \cdot \mathbf{T}_g + \sum_j^{l,e,i} \mathbf{f}_{jg} \tag{2.2b}$$
$$\frac{\partial \alpha_g \rho_g E_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g H_g = -\nabla \cdot \mathbf{q}_g'' + \nabla \cdot [\mathbf{T}_g \cdot \mathbf{u}_g] + \sum_j^{l,e,i} q_{jg}'''. \tag{2.2c}$$

For charge particles, formulae relating physical quantities are more conveniently expressed using number rather than mass density ρ, as many electromagnetic quantities and plasma properties are calculated using number density n and charge density en. As a result, momentum density is written as $m_j n_j \mathbf{u}_j$, and energy density as $n_j \mathcal{E}_j$. The electron conservation equations are

$$\frac{\partial \alpha_g n_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e = 0 \tag{2.3a}$$
$$m_e \left[\frac{\partial \alpha_g n_e \mathbf{u}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \otimes \mathbf{u}_e \right] = -\nabla p_e + \nabla \cdot \mathbf{T}_e + \sum_j^{g,i} \mathbf{f}_{je} \tag{2.3b}$$
$$\frac{\partial \alpha_g n_e \mathcal{E}_e}{\partial t} + \nabla \cdot \alpha_g \rho_e \mathbf{u}_e \mathcal{H}_e = -\nabla \cdot \mathbf{q}_e'' + \nabla \cdot [\mathbf{T}_e \cdot \mathbf{u}_e] + \sum_j^{g,i} q_{je}'''. \tag{2.3c}$$

Finally, the ion conservation equations are

$$\frac{\partial \alpha_g n_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i = 0 \tag{2.4a}$$
$$m \left[\frac{\partial \alpha_g n_i \mathbf{u}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \otimes \mathbf{u}_i \right] = -\nabla p_i + \nabla \cdot \mathbf{T}_i + \sum_j^{g,e} \mathbf{f}_{ji} \tag{2.4b}$$
$$\frac{\partial \alpha_g n_i \mathcal{E}_i}{\partial t} + \nabla \cdot \alpha_g \rho_i \mathbf{u}_i \mathcal{H}_i = -\nabla \cdot \mathbf{q}_i'' + \nabla \cdot [\mathbf{T}_i \cdot \mathbf{u}_i] + \sum_j^{g,e} q_{ji}'''. \tag{2.4c}$$

2.1.2 Relaxation terms

This section details the relaxation processes between the liquid and gas phases. For details about relaxations of charge particles, see Section 2.4.2.

2.1.3 Reaction terms

Reaction and phase change are used interchangeably in this document, since the same chemical species of different states are treated as different fluids. A reaction rate density Γ_r^j describes how many reactions occurs per unit volume per unit time. The subscript r indicates the reaction type and the superscript j indicates the phase of interest that is involved in said reaction. The term is negative for reactants and positive for products. In a binary reaction – for example, boiling – the sum of the two Γ terms is 0; therefore, $\Gamma^l_{\text{boil}} = -\Gamma^g_{\text{boil}}$. Ternary reactions such as ionization and recombination will see an imbalance of particles following a reaction. Section 2.4.1 describes the most commonly found ionization reactions encountered in the system.

Reaction also transfer a portion of the reactant's momentum and energy into the product pool, and the particle momentum $m_j\mathbf{u}_j$ and energy $\mathcal{E}_j$ being transferred are those of the reactants. When expressing this transfer under a summation operator, e.g., $\sum_r \Sigma_r^j \mathcal{E}_r$, Σ_r^j indicates all reactions r that produce particle j, and $\mathcal{E}_r$ means the energy of the reactant in each reaction. Note that this only applies readily for heavy particles; electrons typically involve reaction-specific transfer terms that are described in Section 2.4.1.

If it is convenient to express reaction rate density in mass-based units, the conversion factor is the mass of the particle of interest, m.

2.1.4 Transport terms

The diffusive terms in the momentum and energy equations are $\nabla \cdot \mathbf{T}$ and $\nabla \cdot \mathbf{q}''$, respectively. For neutral particles, the deviatoric stress tensor $\mathbf{T}$ is

$$\mathbf{T} = \mu \left[\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right], \tag{2.5}$$

and the isotropic heat flux vector $\mathbf{q}''$ is

$$\mathbf{q}'' = -\kappa \nabla T. \tag{2.6}$$

The viscosity μ and thermal conductivity κ are given by a material property library. Transport properties of charged particles are described in Section 2.5.

2.1.5 Closure relations

For each set of conservation equations, a closer relation is required to close the system. It usually takes form of an equation of state and relates a conserved variable to a set of primitive variables. Species in the gas phase, neutral or charged, follow an ideal gas assumption.

2.2 Electromagnetic effects

2.2.1 Maxwell's equations

Hydrocarbon materials such that one used in this project can be reasonably assumed to be isotropic and homogeneous. This simplifies the Maxwell's equations in matter because it allows two vectors that are difficult to compute, namely the polarization vector $\mathbf{P}$ and the magnetization vector $\mathbf{M}$, to be parallel to the electric and magnetic fields. Moreover, $\mu_r \approx 1$ is assumed since hydrocarbon are non-magnetic materials. This leads to the following form of Maxwell's equations in matter:

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon_r \epsilon_0} \tag{2.7a}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.7b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.7c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}_f + \mu_0 \epsilon_r \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.7d}$$

In plasma, the free charge density ρ_f and current density $\mathbf{j}_f$ can be easily computed using electron and ion information:

$$\rho_f \equiv e(n_i - n_e) \tag{2.8a}$$

$$\mathbf{j}_f \equiv e(n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.8b}$$

Maxwell's Equations (Equation 2.7) are over-determined; there are 8 scalar equations that only describes 6 scalar variables (the components of $\mathbf{E}$ and $\mathbf{B}$). However, it can be shown that taking the divergence of Equations 2.7c and 2.7d will lead to Equations 2.7b and 2.7a, respectively. Therefore, only Equations 2.7c and 2.7d are necessary to model the $\mathbf{E}$ and $\mathbf{B}$ fields.

2.2.2 Lorentz force

The Lorentz force acts upon a distribution of charged particles has the form

$$\mathbf{f}_j = \pm e n_j \left(\mathbf{E} + \mathbf{u}_j \times \mathbf{B} \right). \tag{2.9}$$

Since electrons are extremely lightweight, its acceleration due to the Lorentz force easily dominates any diffusive or gravitational effects, so they are safe to neglect. The case for ions is a little bit more nuanced; however, the Lorentz force is still expected to be the main momentum contributor.

Electromagnetic waves themselves carry energy, and therefore capable of doing work on a charged system. According to Equation 2.9, the magnetic field does no work on charged particles as its force component is always perpendicular to the charge velocity. The rate of work done by the electric field is then

$$W_j = \mathbf{f}_j \cdot \mathbf{u}_j = \pm e n_j \mathbf{u}_j \cdot \mathbf{E}, \tag{2.10}$$

and should be equal to the rate of energy gained by the particles. This accounts for the change in kinetic energy of a charged particles under the influence of the electric fields.

2.3 Laser interactions

While the model assumes the droplet is at least partially ionized, the plasma state is not formed without a laser source. The laser introduces a fourth particle – the photon – into the system, and it is quickly absorbed by either bound or free electrons. Whereas the term "electrons" has typically referred to free electrons for much of this chapter, in this context it collectively refers to both free and bound electrons.

2.3.1 Energy deposition

This model assumes the laser beam is a line heat source within each droplet even if the laser cross-sectional area is known. The volumetric heat generation, as a result, contains a Dirac delta function $\delta(\boldsymbol{r}_{\text{laser}})$ where $\boldsymbol{r}_{\text{laser}}$ is the laser trajectory. Finite-volume methods blend well with this assumption since the delta function is always integrated out during the formulation.

$$\iiint_{\Gamma \supset \boldsymbol{r}_0} \delta(\boldsymbol{r}_0) dV = 1 \tag{2.11}$$

Therefore, the total energy deposited in a cell that the laser passes through is the difference in laser energy at the cell entrance and exit. If the entrance energy upon cell k is $I_k A$, it will have been attenuated by a factor of $e^{-\alpha_k s_k}$ when exitting the cell – s_k being the distance that the laser beam travels within cell k. Therefore,

$$\iiint_{V_k} q'''_{\text{laser}} dV = I_k A(1 - e^{-\alpha_k s_k}) \tag{2.12}$$

Bound and free electrons differ in how the energy is spent. A free electron, upon absorbing a photon, simply increases in kinetic energy. A bound electron, however, can either be excited or removed from its orbit altogether. If excitation occurs, collisions with the atom causes the electron to rapidly relax and transfer the acquired energy into thermal energy of the bulk liquid; therefore, on the timescale of interest, it is responsible to assume that the energy goes directly into heating the neutral fluid.

With two channels in which laser energy can be deposited, the total attenuation coefficient α is the sum of the bound and free counterparts.

$$\alpha = \alpha_b + \alpha_{\text{ib}}, \tag{2.13}$$

and the fraction of energy that goes into each channel is proportional to the relative magnitude of the two absorption coefficients

$$q_b = I_0 A(1 - e^{-\alpha s})\frac{\alpha_b}{\alpha} \tag{2.14a}$$

$$q_{\gamma e} = I_0 A(1 - e^{-\alpha s})\frac{\alpha_{\text{ib}}}{\alpha}. \tag{2.14b}$$

The subscript "ib" in Equation 2.13 refers to inverse bremsstrahlung, a process in which free electrons, in the vincinty of an ion, absorbs energy and starts to accelerate. The double subscript γe infers as if the heat source were due to a "collision" between a photon and a free electron, similar to how other heat sources are denoted in Section 2.4.2. Inverse bremsstrahlung should take over as the dominant absorption process once a sufficiently opaque medium is formed. The attenuation coefficient will increase proportionally to the inverse bremsstrahlung frequency ν_{ib}

$$\alpha_{\text{ib}} = \frac{\nu_{\text{ib}}\eta}{c} = \frac{\omega_e^2}{\omega^2 + \nu_{ei}^2}\frac{\nu_{ei}\eta}{c}, \tag{2.15}$$

where ω_e is the plasma electron frequency and ω is the laser frequency. The former is defined as

$$\omega_e^2 \equiv \frac{n_e e^2}{\varepsilon_0 m_e}. \tag{2.16}$$

When $\omega \gg \nu_{ei}$, ν_{ib} is usually given in the form

$$\nu_{\text{ib}} = \frac{n_e}{n_c}\nu_{ei}, \tag{2.17}$$

and the critical density n_c is defined in terms of ω

$$\omega^2 = \frac{n_c e^2}{\varepsilon_0 m_e}. \tag{2.18}$$

2.3.2 Photoionization

Photoionization can only occur if the photon energy, $h\nu$, exceeds the binding energy $\mathcal{E}_b$. This is simply the photoelectric effect, and it is typically caused by photons in the ultraviolet regime. For a quantum yield $\Phi \in [0, 1]$ that accounts for the number of photoelectrons produced per incident photon, the photoionization rate density is

$$\Gamma_{\mathrm{pi}} = \Phi \frac{q_b'''}{h\nu}, \tag{2.19}$$

where q_b''' is related to q_b in Equation 2.14a as

$$\iiint q_b''' dV = q_b. \tag{2.20}$$

A newly freed electron will carry an excess energy of $h\nu - \mathcal{E}_b$, which will be deposited towards the energy of all free electrons with a rate of

$$q_{\mathrm{pi}}''' = (h\nu - \mathcal{E}_b)\Gamma_{\mathrm{pi}}. \tag{2.21}$$

Any absorption that do not result in ionization will instead excite bound electrons and serve as a heat source for the neutral particles with a rate of

$$q_{\gamma n}''' = (1 - \Phi) q_b''' = q_b''' - h\nu\Gamma_{\mathrm{pi}}. \tag{2.22}$$

It is possible for a bound electron to simultaneously absorb multiple photons, a phenomenon known as multiphoton ionization [6]. This instead becomes a strongly nonlinear process, and the probability of multiple photons being absorbed by the same electron increases strongly with laser intensity. The minimum number of photons needed to completely remove an electron from its orbit is

$$N \equiv \left\lceil \frac{\mathcal{E}_b}{h\nu} \right\rceil. \tag{2.23}$$

The N-photon ionization rate is proportional to the N-th power of a particular measure of photon flux. In [7], that measure is the photon scalar flux, $\phi = \dfrac{I}{h\nu}$. However, since the laser intensity monotonic decreases with distance travelled, it is better to calculate the ionization rate using a path-averaged scalar flux:

$$\begin{aligned} \overline{\phi} &= \frac{\overline{I}}{h\nu} && (2.24\text{a}) \\ &= \frac{1}{h\nu s} \int_0^s I_0 e^{-\alpha s'} ds' && (2.24\text{b}) \\ &= \frac{I_0}{h\nu} \frac{1 - e^{-\alpha s}}{\alpha s} && (2.24\text{c}) \end{aligned}$$

The rate is then given by

$$\Gamma_{\mathrm{pi}} = \min\left(S_N \overline{\phi}^N n, \frac{q_b'''}{N h\nu} \right) \tag{2.25}$$

Here, S_N is the N-photon ionization rate coefficient and n is the neutral particle number density. The min function is to ensure that the actual ionization rate does not exceed the photon deposition rate. In order for Γ_{pi} to have the correct unit, the unit of S_N is $\mathrm{m}^{2N}\ \mathrm{s}^{N-1}$. If Equation 2.25 were to be in terms of another unit (for example, laser intensity) instead, then the unit of S_N must be changed accordingly. The term N-photon ionization cross section and the symbol σ_N are often used in literature to describe this coefficient [6] [7]; this nomenclature is not adopted here in order to ensure that all cross sections have units of area.

The rate of energy deposited into the neutrals and electrons are

$$q'''_{\mathrm{pi}} = (Nh\nu - \mathcal{E}_b)\Gamma_{\mathrm{pi}} \tag{2.26a}$$
$$q'''_{\gamma n} = q'''_b - Nh\nu\Gamma_{\mathrm{pi}}. \tag{2.26b}$$

For simplicity, bound electrons are assumed to only absorb N photons prior to removal from its parent nucleus. Above-threshold ionization, a phenomenon where an electron absorbs more than N photons, is also possible, but will be neglected here.

2.4 Collisions in plasmas

Species in a plasma exchange their mass, momentum, and energy primarily through local collisions. The collisions described in this section are assumed to be entirely binary; i.e. involving two particles at a time. There are two interactions that can follow a local collision of unlike particles, namely reaction and relaxation. Reactions convert a neutral particles into ions or vice versa, and relaxations drive the system towards equilibrium in absence of external sources.

2.4.1 Reactions

When a free electron collides with a bound electron and departs sufficient energy on it, the latter can be removed from its orbit. Known as collisional ionization, this process is responsible for rapid increase of the free electron population. Not all free electrons can cause collisional ionization; it must have a kinetic energy $\mathcal{E}_e$ that exceeds the binding energy $\mathcal{E}_b$ of the bound electron. Under the fluid model of plasma, electrons in any control volume is modeled as having a single particle-averaged energy. In this section, this averaged energy is denoted $\frac{3}{2}k_BT_e$ in this section as opposed to $\mathcal{E}_e$, which instead represents the energy of an individual electron. Collisional ionization still occurs in a volume where $\frac{3}{2}k_BT_e < \mathcal{E}_b$ since there exists a fraction of electron with a sufficient $\mathcal{E}_e$.

Assuming a Maxwellian distribution of velocity, the probability density function (PDF) of electron speed $f(u_e)$ is [5]

$$f(u_e) = \left(\frac{m_e}{2\pi k_BT_e}\right)^{3/2} 4\pi u_e^2 \exp\frac{-\frac{1}{2}m_eu_e^2}{k_BT_e}. \tag{2.27}$$

To convert Equation 2.27 into a PDF in kinetic energy, the following identity is employed

$$f(\mathcal{E}_e)\left|d\mathcal{E}_e\right| = f(u_e)\left|du_e\right|. \tag{2.28}$$

which requires the derivative

$$\frac{d\mathcal{E}_e}{du_e} = m_eu_e \tag{2.29a}$$
$$\frac{du_e}{d\mathcal{E}_e} = \frac{1}{m_eu_e} = \frac{1}{\sqrt{2m_e\mathcal{E}_e}} \tag{2.29b}$$

The energy-PDF is therefore

$$f(\mathcal{E}_e) = f(u_e)\left|\frac{du_e}{d\mathcal{E}_e}\right| \tag{2.30a}$$

$$= \frac{2}{\sqrt{\pi}}\sqrt{\frac{\mathcal{E}_e}{(k_B T_e)^3}}\exp\frac{-\mathcal{E}_e}{k_B T_e} \tag{2.30b}$$

The electron-impart ionization cross section is calculated using the Binary-Encounter-Bethe (BEB) model [8]:

$$\sigma_{\text{BEB}} = \frac{4\pi a_0^2 r^2}{t+u+1}\left[\frac{\ln t}{2}\left(1-\frac{1}{t^2}\right)+1-\frac{1}{t}-\frac{\ln t}{t+1}\right], \tag{2.31a}$$

where the following dimensionless energies are used

$$r \equiv \frac{\mathcal{R}}{\mathcal{E}_b} \tag{2.31b}$$

$$t \equiv \frac{\mathcal{E}_e}{\mathcal{E}_b} \tag{2.31c}$$

$$u \equiv \frac{\mathcal{U}}{\mathcal{E}_b} \tag{2.31d}$$

Let Γ_r be the ionization rate density of a reaction r, which has a positive value if ions are created and negative if they are consumed. For collisional ionization, the rate density is

$$\Gamma_{\text{ci}} = S_{\text{ci}} n_e n \tag{2.32a}$$

$$S_{\text{ci}} \equiv \int_{\mathcal{E}_b}^{\infty} \sigma_{\text{BEB}}(\mathcal{E}_e) v(\mathcal{E}_e) f(\mathcal{E}_e) d\mathcal{E}_e. \tag{2.32b}$$

The rate constant S_{ci} has units of volumetric flow rate (m^3/s).

The reverse process of ionization is recombination, where an ion captures a free electron to become neutral. Recombination, fortunately, does not have an energy threshold, and the reaction rate density is

$$\Gamma_{\text{rec}} = -S_{\text{rec}} n_e n_i. \tag{2.33}$$

Reactions do not just alter the amount of each species in a plasma; it also transfer a portion of the reactant's momentum and energy directly into the product, assuming no enthalpy of reaction. These transfers are typically more important for ions and neutrals than they are for electrons, as the latter have other sources that contribute to their momentum and energy a lot more significantly. Only account for the transfer of conservative variables in reactive processes, the conservation equations read

$$\frac{\partial n_i}{\partial t} = -\frac{\partial n}{\partial t} = \sum_r \Gamma_r \tag{2.34a}$$

$$\frac{\partial n_i \mathbf{u}_i}{\partial t} = -\frac{\partial n \mathbf{u}}{\partial t} = \sum_r \Gamma_r \mathbf{u}_r \tag{2.34b}$$

$$\frac{\partial n_i \mathcal{E}_i}{\partial t} = -\frac{\partial n \mathcal{E}}{\partial t} = \sum_r \Gamma_r \mathcal{E}_r, \tag{2.34c}$$

where the subscript r on a state variable indicates that the variable belongs to the reactant species.

2.4.2 Relaxations

A collision between a charged particle j and a neutral particle results in a force roughly equal to

$$\mathbf{f}_{jn} = \frac{m_j n_j(\mathbf{u}_j - \mathbf{u})}{\tau_{jn}}. \tag{2.35}$$

Here the double subscript jn refers to a transfer from a particle j to a neutral particle. By Newton's third law of motion, $\mathbf{f}_{jn} = -\mathbf{f}_{nj}$. The mean time between collisions τ_{jn} is approximately constant and is the reciprocal of the momentum-transfer collision frequency

$$\nu_{jn}^M \equiv \frac{1}{\tau_{jn}} = n\sigma_{jn}^M \|\mathbf{u}_j - \mathbf{u}\|, \tag{2.36}$$

with σ_{jn}^M denotes the equivalent cross section of momentum transfer between j and n. It is typically a function of charged particle speed; the average value over all velocities in a Maxwellian distribution is typically used in Equation 2.36.

Collisions between two charged particles are Coulomb collisions mediated by the electric field. Like-particle (electron-electron or ion-ion) collisions are often neglected because they preserve the momentum and energy between themselves, and they contribute very little to diffusion. This is in contrast with electron-ion collisions, where there exists a net transfer of momentum (and energy) between the two particles. In a similar formulation as Equation 2.35, the force exerted between an ion and an electron during their collision is

$$\mathbf{f}_{ei} = m_e n_e(\mathbf{u}_i - \mathbf{u}_e)\nu_{ei}^M, \tag{2.37}$$

and the electron-ion momentum-transfer collision frequency ν_{ei}^M is

$$\nu_{ei}^M = C\frac{n_i Z^2 e^4}{(4\pi\varepsilon_0)^2\sqrt{m_e(k_B T_e)^3}}\ln\Lambda_e, \tag{2.38}$$

where C is a prefactor on the order of $\mathcal{O}(1)$ that differs depending on the text. Chen gives $C = \pi$, while the FLASH code uses $C = \frac{4}{3}\sqrt{2\pi}$ [2][3]. The latter will be used in this model. The Coulomb logarithm $\ln\Lambda$ is a correction factor that accounts for the cumulative effect of many small-angle deflections in a collision and is given by

$$\ln\Lambda_e \equiv \ln\left[\frac{4\pi(\varepsilon_0 k_B T_e)^{3/2}}{Ze^3\sqrt{n_e}}\right]. \tag{2.39}$$

In both Equations 2.38 and 2.39, the average ionization number Z can be assumed to be 1.

For all types of collisions, the energy loss due to relaxation term can be expressed as a heat source that flows from particle j to j'

$$q_{jj'}''' = -q_{j'j}''' = \frac{3}{2}n_j\nu_{jj'}^K k_B(T_j - T_{j'}). \tag{2.40}$$

The energy-transfer collision frequency ν^K is related to its momentum counterpart ν^M by

$$\frac{\nu_{jj'}^K}{\nu_{jj'}^M} = \frac{2m_j}{m_j + m_{j'}}. \tag{2.41}$$

The index j in Equation 2.41 refers to the lighter particle out of the pair, so it is always the electron if it is involved in a collision, and the ion in a ion-neutral collision. In the former case, $\nu_{ej'}^K \ll \nu_{ej'}^M$, which means energy is relaxed at a much slower rate than momentum is. Physically, upon colliding with a heavy particle, an electron is frequently scattered before a significant portion of its kinetic energy is lost. In an ion-neutral collision, due to the near-identical mass of the two particles, energy and momentum are relaxed on the same timescale – i.e., $\nu_{in}^K = \nu_{in}^M$.

2.5 Charged particle transport

Transport of neutral particles are caused by the "random walk" movement of the particles, and because there is no inherent preference for a particular direction, the transported quantity – being momentum or energy – is eventually equilibrated. This is also true in the case of charged particles in the *unmagnetized* limits; that is when $\|\mathbf{B}\| = 0$. The existence of a magnetic field gives rise to anisotropy in heat conduction, since the particles are forced to spiral along the magnetic field lines, so the rate of transport is the highest in the direction parallel to the magnetic field lines, i.e. $\hat{\mathbf{b}} = \dfrac{\hat{\mathbf{B}}}{\|\mathbf{B}\|}$. On the other hand, motion across the field lines (the perpendicular direction) are strongly suppressed, so there is little transport in that direction. The other direction, which is perpendicular to both $\hat{\mathbf{b}}$ and the gradient of the transported quantity and termed *cross* direction in this document, is the result of the Righi–Leduc effect.

Transport processes in plasmas are pioneered by Braginskii in his 1965 article [1], where the transport properties – namely viscosity and thermal conductivity – are given as function of magnetization χ (also called the Hall parameter). It is defined as

$$\chi_j \equiv \frac{\omega_{\mathrm{ce},j}}{\nu_{ji}},\ j = e, i \tag{2.42}$$

and the electron or ion gyrofrequency is

$$\omega_{\mathrm{ce},j} \equiv \frac{eB}{m_j}. \tag{2.43}$$

The electron-ion collision frequency is given in Equation 2.38, and the ion-ion collision frequency is

$$\nu_{ii} = \frac{4}{3}\sqrt{\pi}\frac{n_i Z^4 e^4}{(4\pi\varepsilon_0)^2\sqrt{m(k_B T_i)^3}} \ln\Lambda_i. \tag{2.44}$$

2.5.1 Heat flux

The electron heat flux has a conductive part in terms of the temperature gradient ∇T_e and a thermoelectric part in terms of the current density $\mathbf{j}$. Both ∇T_e and $\mathbf{j}$ are each broken down into three components, as followed:

$$\mathbf{q}''_e = -\kappa^e_\parallel \nabla_\parallel T_e - \kappa^e_\perp \nabla_\perp T_e - \kappa^e_\times \nabla_\times T_e - \frac{k_B T_e}{e}\left(\beta_\parallel \mathbf{j}_\parallel + \beta_\perp \mathbf{j}_\perp + \beta_\times \mathbf{j}_\times\right) \tag{2.45}$$

The three components of ∇T_e are

$$\nabla_\parallel T_e = (\hat{\mathbf{b}} \cdot \nabla T_e)\hat{\mathbf{b}} \tag{2.46a}$$
$$\nabla_\perp T_e = \nabla T_e - \nabla_\parallel T_e \tag{2.46b}$$
$$\nabla_\times T_e = \hat{\mathbf{b}} \times \nabla T_e, \tag{2.46c}$$

and the corresponding components of $\mathbf{j}$ are defined similarly.

Both the conductive and the thermoelectric contributions in Equation 2.45 can be rewritten compactly in terms of their respective original vectors, ∇T_e and $\mathbf{j}$. Take the conductive part, for example,

$$\mathbf{q}''_{\mathrm{cond}} = -\kappa^e_\parallel(\hat{\mathbf{b}} \cdot \nabla T_e)\hat{\mathbf{b}} - \kappa^e_\perp(\nabla T_e - (\hat{\mathbf{b}} \cdot \nabla T_e)\hat{\mathbf{b}}) - \kappa^e_\times \hat{\mathbf{b}} \times \nabla T_e \tag{2.47a}$$
$$= -\underbrace{\left(\kappa^e_\perp \mathbf{I} + (\kappa^e_\parallel - \kappa^e_\perp)\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \kappa^e_\times \mathbf{B}_\times\right)}_{\mathbf{K}_e} \nabla T_e, \tag{2.47b}$$

where $\mathbf{B}_\times$ is the antisymmetric matrix

$$\mathbf{B}_\times = \begin{bmatrix} 0 & -b_z & b_y \\ b_z & 0 & -b_x \\ -b_y & b_x & 0 \end{bmatrix}, \tag{2.48}$$

that satisfies $\mathbf{B}_\times \mathbf{v} = \hat{\mathbf{b}} \times \mathbf{v}$ for any vector $\mathbf{v}$.

Similarly, the thermoelectric contribution of the electron heat flux can be written as

$$\mathbf{q}''_{\text{thermo}} = -\underbrace{\frac{k_B T_e}{e}\left(\beta_\perp \mathbf{I} + (\beta_\parallel - \beta_\perp)\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \beta_\times \mathbf{B}_\times\right)}_{\mathbf{\Pi}}\mathbf{j}, \tag{2.49}$$

so that $\mathbf{q}''_e = -\mathbf{K}_e \nabla T_e - \mathbf{\Pi}\mathbf{j}$.

The thermal conductivities are given in Epperlein and Haines (1986) (which are corrections of the Braginskii coefficients) as rational functions of χ_e, and their polynomial coefficients are given in Table IV [4]:

$$\kappa^e_\parallel = \hat{\kappa}_e \frac{\gamma'_0}{c'_0}, \tag{2.50a}$$

$$\kappa^e_\perp = \hat{\kappa}_e \frac{\gamma'_1 \chi_e + \gamma'_0}{\chi_e^3 + c'_2 \chi_e^2 + c'_1 \chi_e + c'_0}, \tag{2.50b}$$

$$\kappa^e_\times = \hat{\kappa}_e \frac{\chi_e(\gamma''_1 \chi_e + \gamma''_0)}{\chi_e^3 + c''_2 \chi_e^2 + c''_1 \chi_e + c''_0}, \tag{2.50c}$$

with the prefractor $\hat{\kappa}_e$ defined as

$$\hat{\kappa}_e \equiv \frac{n_e k_B T_e}{m_e \nu^M_{ei}}. \tag{2.50d}$$

Also from Epperlein and Haines, the thermoelectric coefficients are given as the following rational functions, whose polynomial coefficients are given in Table III [4]:

$$\beta_\parallel = \frac{\beta'_0}{(b'_0)^{8/9}}, \tag{2.51a}$$

$$\beta_\perp = \frac{\beta'_1 \chi_e + \beta'_0}{(\chi_e^3 + b'_2 \chi_e^2 + b'_1 \chi_e + b'_0)^{8/9}}, \tag{2.51b}$$

$$\beta_\times = \frac{\chi_e(\beta''_1 \chi_e + \beta''_0)}{\chi_e^3 + b''_2 \chi_e^2 + b''_1 \chi_e + b''_0}. \tag{2.51c}$$

In the unmagnetized limit ($\chi_e \to 0$), $\kappa^e_\parallel = \kappa^e_\perp = \kappa_e$ and $\beta_\parallel = \beta_\perp = \beta$ while all cross terms vanish. The isotropic flux is recovered:

$$\mathbf{q}''_e = -\kappa_e \nabla T_e - \frac{\beta k_B T_e}{e}\mathbf{j} \tag{2.52}$$

Ion transport follows a simpler formulation where the thermoelectric contribution is neglected, leaving only the conduction term:

$$\mathbf{q}''_i = -\left(\kappa^i_\perp \mathbf{I} + (\kappa^i_\parallel - \kappa^i_\perp)\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \kappa^i_\times \mathbf{B}_\times\right)\nabla T_i. \tag{2.53}$$

The ion conductivities follow the identical rational form as the electron counterparts in Equation 2.50, but with χ_i being the magnetization and a different prefactor

$$\hat{\kappa}_i \equiv \frac{n_i k_B T_i}{m \nu_{ii}}. \tag{2.54}$$

Ion thermal conductivity is expected to be much lower than that of electrons.

2.5.2 Momentum flux

Braginskii gives the total stress tensor as the sum of five component tensors:

$$\mathbf{T} = \sum_{n=0}^{4} \mathbf{T}_n \tag{2.55a}$$

$$\mathbf{T}_0 = 3\mu_0 \left(\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \frac{1}{3}\mathbf{I} \right) (\hat{\mathbf{b}} \cdot \mathbf{W}\hat{\mathbf{b}}) \tag{2.55b}$$

$$\mathbf{T}_1 = \mu_1 \left[\mathbf{PWP} + \frac{1}{2}\mathbf{P}(\hat{\mathbf{b}} \cdot \mathbf{W}\hat{\mathbf{b}}) \right] \tag{2.55c}$$

$$\mathbf{T}_2 = \mu_2 \left[\mathbf{PW}\hat{\mathbf{b}}\hat{\mathbf{b}}^T + \hat{\mathbf{b}}\hat{\mathbf{b}}^T\mathbf{WP} \right] \tag{2.55d}$$

$$\mathbf{T}_3 = -\frac{\mu_3}{2} \left[\mathbf{B}_\times \mathbf{WP} - \mathbf{PWB}_\times \right] \tag{2.55e}$$

$$\mathbf{T}_4 = -\mu_4 \left[\mathbf{B}_\times \mathbf{W}\hat{\mathbf{b}}\hat{\mathbf{b}}^T - \hat{\mathbf{b}}\hat{\mathbf{b}}^T\mathbf{WB}_\times \right], \tag{2.55f}$$

where the projection matrix $\mathbf{P}$ is

$$\mathbf{P} \equiv \mathbf{I} - \hat{\mathbf{b}}\hat{\mathbf{b}}^T, \tag{2.55g}$$

and the rate-of-strain tensor $\mathbf{W}$ is

$$\mathbf{W} \equiv \left[\nabla\mathbf{u} + (\nabla\mathbf{u})^T \right] - \frac{2}{3}\mathbf{I}\nabla \cdot \mathbf{u} \tag{2.55h}$$

The viscosities are rational functions of χ, and their polynomial coefficients are given in Equations 4.44 through 4.45 of Braginskii [1]:

$$\mu_0^j = \hat{\mu}_j \frac{\alpha_0^{j'}}{a_0^j} \tag{2.56a}$$

$$\mu_2^j = \hat{\mu}_j \frac{\alpha_2^{j'}\chi_j^2 + \alpha_0^{j'}}{\chi_j^4 + a_2^j\chi_j^2 + a_0^j} \tag{2.56b}$$

$$\mu_4^j = \hat{\mu}_j \frac{\chi_j(\chi_j^2 + \alpha_0^{j''})}{\chi_j^4 + a_2^j\chi_j^2 + a_0^j} (\delta_{ji} - \delta_{je}) \tag{2.56c}$$

$$\mu_1^j = \mu_2^j(2\chi_j) \tag{2.56d}$$

$$\mu_3^j = \mu_4^j(2\chi_j), \tag{2.56e}$$

with the leading prefractor $\hat{\mu}_j$ defined as

$$\hat{\mu}_j \equiv \frac{n_j k_B T_j}{\nu_{ji}^M}. \tag{2.56f}$$

As a note, the term $(\delta_{ji} - \delta_{je})$ evaluates to 1 for ions and -1 for electrons.

In the unmagnetized limit ($\chi_e \to 0$), the gyroviscous terms $\mathbf{T}_3$ and $\mathbf{T}_4$ vanish, while $\mu_0 = \mu_1 = \mu_2 = \mu$, and the stress tensor becomes

$$\mathbf{T} = \mu\mathbf{W}, \tag{2.57}$$

which is identical to the conventional deviatoric stress tensor.

2.6 Summary

The system of equations have a total of 27 scalar variables, organized into five categories. The liquid phase equations are

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot \alpha_l \mathbf{u}_l = \mathcal{P}_{gl} \tag{2.58}$$

$$\frac{\partial \alpha_l \rho_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l = -\Sigma_{\text{evap}} \tag{2.59}$$

$$\frac{\partial \alpha_l \rho_l \mathbf{u}_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l \otimes \mathbf{u}_l = -\nabla p_l + \nabla \cdot \mathbf{T}_l + \rho_l \mathbf{g} + \mathbf{f}_{gl} \tag{2.60}$$

$$\frac{\partial \alpha_l \rho_l E_l}{\partial t} + \nabla \cdot \alpha_l \rho_l \mathbf{u}_l H_l = -\nabla \cdot \mathbf{q}_l'' + \nabla \cdot [\mathbf{T}_l \cdot \mathbf{u}_l] + \sum_j^{g,\gamma} q_{jl}''' \tag{2.61}$$

The gas phase equations are

$$\frac{\partial \alpha_g \rho_g}{\partial t} + \nabla \cdot \rho_g \mathbf{u}_g = \frac{M}{N_A} \sum_r \Gamma_r^g \tag{2.62}$$

$$\frac{\partial \alpha_g \rho_g \mathbf{u}_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g \otimes \mathbf{u}_g = -\nabla p_g + \nabla \cdot \mathbf{T}_g + \rho_g \mathbf{g} + \sum_j^{l,e,i} \mathbf{f}_{jg} + \frac{M}{N_A} \sum_r \Gamma_r^g \mathbf{u}_r \tag{2.63}$$

$$\frac{\partial \alpha_g \rho_g E_g}{\partial t} + \nabla \cdot \alpha_g \rho_g \mathbf{u}_g H_g = -\nabla \cdot \mathbf{q}_g'' + \nabla \cdot [\mathbf{T}_g \cdot \mathbf{u}_g] + \sum_j^{l,e,i,\gamma} q_{jg}''' + \frac{M}{N_A} \sum_r \Gamma_r^g E_r \tag{2.64}$$

The electron equations are

$$\frac{\partial \alpha_g n_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e = \sum_r \Gamma_r^e \tag{2.65}$$

$$m_e \left(\frac{\partial \alpha_g n_e \mathbf{u}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \otimes \mathbf{u}_e \right) = -\nabla p_e - e \alpha_g n_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) + \sum_j^{g,i} \mathbf{f}_{je} \tag{2.66}$$

$$\frac{\partial \alpha_g n_e \mathcal{E}_e}{\partial t} + \nabla \cdot \alpha_g n_e \mathbf{u}_e \mathcal{H}_e = -e \alpha_g n_e \mathbf{u}_e \cdot \mathbf{E} + \sum_j^{g,i,\gamma} q_{je}''' + \sum_r \Gamma_r^e (\mathcal{E}_r - \mathcal{E}_b) \tag{2.67}$$

The ion equations are

$$\frac{\partial \alpha_g n_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i = \sum_r \Gamma_r^i \tag{2.68}$$

$$m_i \left(\frac{\partial \alpha_g n_i \mathbf{u}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \otimes \mathbf{u}_i \right) = -\nabla p_i + e \alpha_g n_i (\mathbf{E} + \mathbf{u}_i \times \mathbf{B}) + \sum_j^{g,e} \mathbf{f}_{ji} + m_i \sum_r \Gamma_r^i \mathbf{u}_r \tag{2.69}$$

$$\frac{\partial \alpha_g n_i \mathcal{E}_i}{\partial t} + \nabla \cdot \alpha_g n_i \mathbf{u}_i \mathcal{H}_i = e \alpha_g n_i \mathbf{u}_i \cdot \mathbf{E} + \sum_j^{g,e} q_{ji}''' + \sum_r \Gamma_r^i \mathcal{E}_r \tag{2.70}$$

Finally, Maxwell's Equations are needed to model the electric and magnetic fields.

$$\varepsilon_r \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B} - e\alpha_g (n_i \mathbf{u}_i - n_e \mathbf{u}_e) \tag{2.71}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{2.72}$$

Chapter 3

Numerical methods

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